

Cryptography and Network Security

Fifth Edition
by William Stallings

Chapter2

Based on the lecture slides by Lawrie Brown for “Cryptography and Network Security”, 5/e, by William Stallings, Chapter 2 – “Classical Encryption Techniques”.

Chapter 2 - Classical Encryption Techniques

- *"I am fairly familiar with all the forms of secret writings, and am myself the author of a trifling monograph upon the subject, in which I analyze one hundred and sixty separate ciphers," said Holmes..*
—*The Adventure of the Dancing Men, Sir Arthur Conan Doyle*

Opening quote.

Cryptographic Tools

- ❑ cryptographic algorithms important element in security services
- ❑ review various types of elements
 - symmetric encryption
 - public-key (asymmetric) encryption
 - digital signatures and key management
 - secure hash functions
- ❑ example is use to encrypt stored data

An important element in many computer security services and applications is the use of cryptographic algorithms. This chapter provides an overview of the various types of algorithms, together with a discussion of their applicability. For each type of algorithm, we introduce the most important standardized algorithms in common use. We begin with symmetric encryption, which is used in the widest variety of contexts, primarily to provide confidentiality. Next, we examine secure hash functions and discuss their use in message authentication. The next section examines public-key encryption, also known as asymmetric encryption. We then look in at the two most important applications of public-key encryption, namely digital signatures and key management. In the case of digital signatures, asymmetric encryption and secure hash functions are combined to produce an extremely useful tool. Finally, in this chapter we provide an example of an application area for cryptographic algorithms by looking at the encryption of stored data.

Symmetric Encryption

- ❑ or conventional / private-key / single-key
- ❑ sender and recipient share a common key
- ❑ all classical encryption algorithms are single-key
- ❑ was only type prior to invention of public-key in 1970's
- ❑ and by far most widely used

Symmetric encryption, also referred to as conventional encryption or single-key encryption, was the only type of encryption in use prior to the development of public-key encryption in the 1970s. It remains by far the most widely used of the two types of encryption. All traditional schemes are **symmetric / single key / private-key** encryption algorithms, with a **single key**, used for both encryption and decryption. Since both sender and receiver are equivalent, either can encrypt or decrypt messages using that common key.

Some Basic Terminology

- ❑ **Plaintext** - original message
- ❑ **Ciphertext** - coded message
- ❑ **Encryption algorithm** -algorithm for transforming plaintext to ciphertext (**enciphering**)
- ❑ **Decryption algorithm** -algorithm for transforming ciphertext to plaintext (**deciphering**)
- ❑ **Secret Key** - info used in Encryption/Decryption algorithms known only to sender/receiver

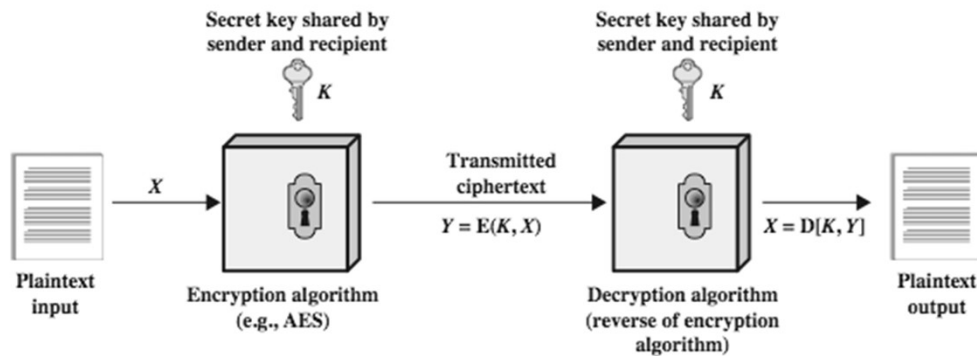
Briefly review some terminology used throughout the course.

Some Basic Terminology

- ❑ **cryptography** - study of encryption schemes/methods. Such scheme is cryptographic system or cipher.
- ❑ **cryptanalysis (codebreaking)** - study of schemes/methods of deciphering ciphertext *without* knowing key
- ❑ **cryptology** - field of both cryptography and cryptanalysis

Briefly review some terminology used throughout the course.

2.1 Symmetric Cipher Model



Detail the five ingredients of the symmetric cipher model, shown in Stallings Figure 2.1:

- plaintext - original message
- encryption algorithm – performs substitutions/transformations on plaintext
- secret key – control exact substitutions/transformations used in encryption algorithm
- ciphertext - scrambled message
- decryption algorithm – inverse of encryption algorithm

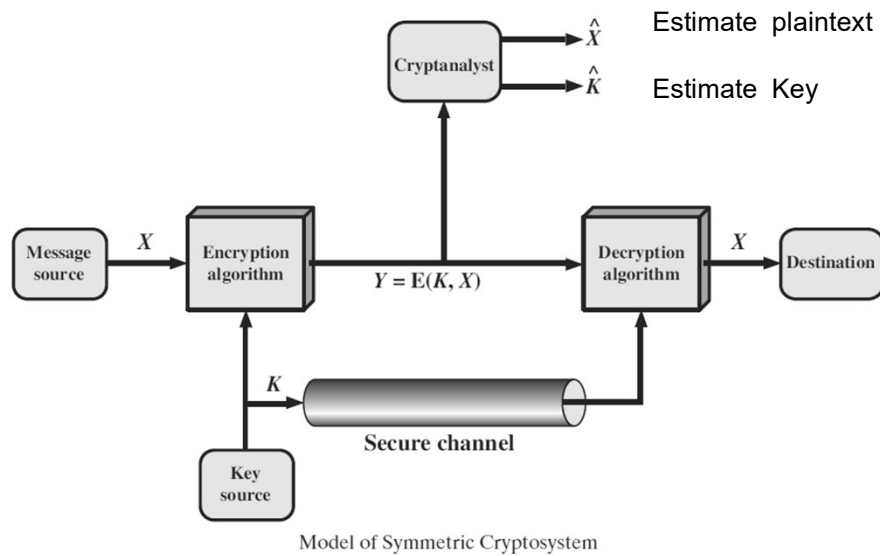
Requirements

- ❑ two requirements for secure use of symmetric encryption:
 - a strong encryption algorithm
 - a secret key known only to sender / receiver
- ❑ mathematically have:
 - $Y = E(K, X)$
 - $X = D(K, Y)$
- ❑ assume encryption algorithm is known
- ❑ implies a secure channel to distribute key

There are two requirements for secure use of conventional encryption that mean we assume that it is impractical to decrypt a message on the basis of the ciphertext plus knowledge of the encryption/decryption algorithm, and hence do not need to keep the algorithm secret; rather we only need to keep the key secret. This feature of symmetric encryption is what makes it feasible for widespread use. It allows easy distribution of s/w and h/w implementations.

Can take a closer look at the essential elements of a symmetric encryption scheme: mathematically it can be considered a pair of functions with: plaintext X , ciphertext Y , key K , encryption algorithm E , decryption algorithm D . The intended receiver, in possession of the key, is able to invert the transformation. An opponent, observing Y but not having access to K or X , may attempt to recover X or K .

Model of symmetric cryptosystem



Cryptography

□ Can characterize cryptographic system by:

- Type of encryption operations used
 - Substitution: each element in the plaintext (bit, letter, group of bits or letters) is mapped into another element.
 - Transposition: elements in the plaintext are rearranged.
 - Product: involve multiple stages of substitutions and transpositions.
- Number of keys used
 - single-key or private
 - two-key or public
- Way in which plaintext is processed
 - block
 - stream

Cryptographic systems can be characterized along these three independent dimensions.

1.The type of operations used for transforming plaintext to ciphertext. All encryption algorithms are based on two general principles: substitution, in which each element in the plaintext (bit, letter, group of bits or letters) is mapped into another element, and transposition, in which elements in the plaintext are rearranged. The fundamental requirement is that no information be lost (that is, that all operations are reversible). Most systems, referred to as product systems, involve multiple stages of substitutions and transpositions.

2.The number of keys used. If both sender and receiver use the same key, the system is referred to as symmetric, single-key, secret-key, or conventional encryption. If the sender and receiver use different keys, the system is referred to as asymmetric, two-key, or public-key encryption.

3.The way in which the plaintext is processed. A block cipher processes the input one block of elements at a time, producing an output block for each input block. A stream cipher processes the input elements continuously, producing output one element at a time, as it goes along.

Cryptanalysis

- ❑ objective to recover key not just message
- ❑ general approaches:
 - cryptanalytic attack
 - brute-force attack
- ❑ if either succeed all key use compromised

Typically objective is to recover the key in use rather than simply to recover the plaintext of a single ciphertext. There are two general approaches:

- **Cryptanalysis:** relies on the nature of the algorithm plus perhaps some knowledge of the general characteristics of the plaintext or even some sample plaintext- ciphertext pairs. This type of attack exploits the characteristics of the algorithm to attempt to deduce a specific plaintext or to deduce the key being used.

- **Brute-force attacks** try every possible key on a piece of ciphertext until an intelligible translation into plaintext is obtained. On average, half of all possible keys must be tried to achieve success.

If either type of attack succeeds in deducing the key, the effect is catastrophic: All future and past messages encrypted with that key are compromised.

Attacking Symmetric Encryption

- ❑ cryptanalysis
 - rely on nature of the algorithm
 - plus some knowledge of plaintext characteristics
 - even some sample plaintext-ciphertext pairs
 - exploits characteristics of algorithm to deduce specific plaintext or key
- ❑ brute-force attack
 - try all possible keys on some ciphertext until get an intelligible translation into plaintext

There are two general approaches to attacking a symmetric encryption scheme. The first attack is known as **cryptanalysis**. Cryptanalytic attacks rely on the nature of the algorithm plus perhaps some knowledge of the general characteristics of the plaintext or even some sample plaintext-ciphertext pairs. This type of attack exploits the characteristics of the algorithm to attempt to deduce a specific plaintext or to deduce the key being used. If the attack succeeds in deducing the key, the effect is catastrophic: All future and past messages encrypted with that key are compromised. The second method, known as the **brute-force attack**, is to try every possible key on a piece of ciphertext until an intelligible translation into plaintext is obtained. On average, half of all possible keys must be tried to achieve success.

Cryptanalytic Attacks

Type of Attack	Known to Cryptanalyst
Ciphertext Only	• Encryption algorithm • Ciphertext
Known Plaintext	• Encryption algorithm • Ciphertext • One or more plaintext–ciphertext pairs formed with the secret key
Chosen Plaintext	• Encryption algorithm • Ciphertext • Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key
Chosen Ciphertext	• Encryption algorithm • Ciphertext • Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key
Chosen Text	• Encryption algorithm • Ciphertext • Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key • Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key

Stallings Table 2.1 summarizes the various types of cryptanalytic attacks, based on the amount of information known to the cryptanalyst, from least to most. The most difficult problem is presented when all that is available is the ciphertext only. In some cases, not even the encryption algorithm is known, but in general we can assume that the opponent does know the algorithm used for encryption. Then with increasing information have the other attacks. Generally, an encryption algorithm is designed to withstand a known-plaintext attack.

Brute Force Search

- ❑ always possible to simply try every key
- ❑ most basic attack, proportional to key size
- ❑ assume either know / recognise plaintext

Key Size (bits)	Number of Alternative Keys	Time required at 1 decryption/ μ s	Time required at 10^6 decryptions/ μ s
32	$2^{32} = 4.3 \times 10^9$	$2^{31} \mu\text{s} = 35.8 \text{ minutes}$	2.15 milliseconds
56	$2^{56} = 7.2 \times 10^{16}$	$2^{55} \mu\text{s} = 1142 \text{ years}$	10.01 hours
128	$2^{128} = 3.4 \times 10^{38}$	$2^{127} \mu\text{s} = 5.4 \times 10^{24} \text{ years}$	$5.4 \times 10^{18} \text{ years}$
168	$2^{168} = 3.7 \times 10^{50}$	$2^{167} \mu\text{s} = 5.9 \times 10^{36} \text{ years}$	$5.9 \times 10^{30} \text{ years}$
26 characters (permutation)	$26! = 4 \times 10^{26}$	$2 \times 10^{26} \mu\text{s} = 6.4 \times 10^{12} \text{ years}$	$6.4 \times 10^6 \text{ years}$

A brute-force attack involves trying every possible key until an intelligible translation of the ciphertext into plaintext is obtained. On average, half of all possible keys must be tried to achieve success. Stallings Table 2.2 shows how much time is required to conduct a brute-force attack, for various common key sizes (DES is 56, AES is 128, Triple-DES is 168, plus general mono-alphabetic cipher), where either a single system or a million parallel systems, are used.

Exhaustive Key Search

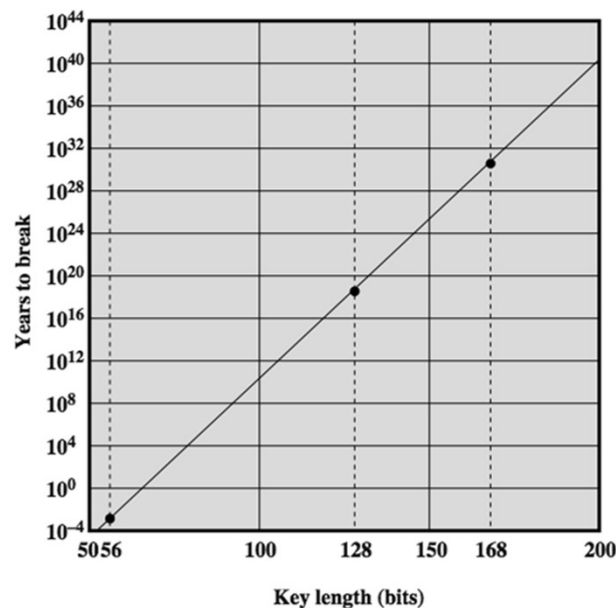


Table 2.1 from the text shows how much time is involved for various key sizes. The table shows results for each key size, assuming that it takes 1 μ s to perform a single decryption, a reasonable order of magnitude for today's computers. With the use of massively parallel organizations of microprocessors, it may be possible to achieve processing rates many orders of magnitude greater. The final column of the table considers the results for a system that can process 1 million keys per microsecond. At this performance level, a 56-bit key is no longer computationally secure.

The assumption of one encryption per microsecond is overly conservative. The widely used encryption scheme, the Data Encryption Standard (DES) was finally and definitively proved insecure in July 1998, when the Electronic Frontier Foundation (EFF) announced that it had broken a DES encryption using a special-purpose "DES cracker" machine that was built for less than \$250,000. The attack took less than three days. The EFF has published a detailed description of the machine, enabling others to build their own cracker [EFF98]. It is important to note that there is more to a key-search attack than simply running through all possible keys. Unless known plaintext is provided, the analyst must be able to recognize plaintext as plaintext. If the message is just plain text in English, then the result pops out easily. If the message is some more general type of data, such as a numerical file, and this has been compressed, the problem becomes even more difficult to automate. Thus, to supplement the brute-force approach, some degree of knowledge about the expected plaintext is needed, and some means of automatically distinguishing plaintext from garble is also needed. The EFF approach addresses this issue as well and introduces some automated techniques that would be effective in many contexts. Figure 2.2 shows how long it would take to crack a DES-style algorithm as a function of key size.

More Definitions

➤ **unconditional security**

- no matter how much computer power or time is available, the cipher cannot be broken since the ciphertext provides insufficient information to uniquely determine the corresponding plaintext

➤ **computational security**

- given limited computing resources (eg time needed for calculations is greater than age of universe), the cipher cannot be broken

Two more definitions are worthy of note. An encryption scheme is unconditionally secure if the ciphertext generated by the scheme does not contain enough information to determine uniquely the corresponding plaintext, no matter how much ciphertext is available. An encryption scheme is said to be computationally secure if either the cost of breaking the cipher exceeds the value of the encrypted information, or the time required to break the cipher exceeds the useful lifetime of the information. Unconditional security would be nice, but the only known such cipher is the **one-time pad** (later).

For all reasonable encryption algorithms, we have to assume computational security where it either takes too long, or is too expensive, to bother breaking the cipher.

Classical Substitution Ciphers

- ❑ where letters of plaintext are replaced by other letters or by numbers or symbols
- ❑ or if plaintext is viewed as a sequence of bits, then substitution involves replacing plaintext bit patterns with ciphertext bit patterns

In this section and the next, we examine a sampling of what might be called classical encryption techniques. A study of these techniques enables us to illustrate the basic approaches to symmetric encryption used today and the types of cryptanalytic attacks that must be anticipated. The two basic building blocks of all encryption technique are substitution and transposition. We examine these in the next two sections. Finally, we discuss a system that combine both substitution and transposition.

A substitution technique is one in which the letters of plaintext are replaced by other letters or by numbers or symbols. If the plaintext is viewed as a sequence of bits, then substitution involves replacing plaintext bit patterns with ciphertext bit patterns.

Caesar Cipher

- ❑ earliest known substitution cipher
- ❑ by Julius Caesar
- ❑ first attested use in military affairs
- ❑ replaces each letter by 3rd letter on
- ❑ example:

meet me after the toga party
PHHW PH DIWHU WKH WRJD SDUWB

Substitution ciphers form the first of the fundamental building blocks. The core idea is to replace one basic unit (letter/byte) with another. Whilst the early Greeks described several substitution ciphers, the first attested use in military affairs of one was by Julius Caesar, described by him in *Gallic Wars* (cf. Kahn pp83-84). Still call any cipher using a simple letter shift a **caesar cipher**, not just those with shift 3.

Caesar Cipher

- can define transformation as:

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

- mathematically give each letter a number

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

- then have Caesar cipher as:

$$c = E(k, p) = (p + k) \bmod (26)$$

$$p = D(k, c) = (c - k) \bmod (26)$$

This mathematical description uses **modulo (clock) arithmetic**. Here, when you reach Z you go back to A and start again. Mod 26 implies that when you reach 26, you use 0 instead (ie the letter after Z, or $25 + 1$ goes to A or 0).

Example: howdy (7,14,22,3,24) encrypted using key f (ie a shift of 5) is MTBID

Cryptanalysis of Caesar Cipher

➤ only have 26 possible ciphers

● A maps to A,B,..Z

➤ could simply try each in turn

➤ a **brute force search**

➤ given ciphertext, just try all shifts of letters

KEY	PHHW	PH	DIWHU	WKH	WRJD	SDUWB
1	oggv	og	chvgt	vjg	vqic	rcuva
2	nffu	nf	bgufs	uif	uphb	qbsuz
3	meet	me	after	the	toga	party
4	ldds	ld	zesdq	sgd	snfz	ozqsx
5	kccr	kc	ydrop	rfc	rmey	nyprw
6	jbbq	jb	xcqbo	qeb	qldx	mxoqv
7	iaap	ia	wbpan	pda	pkcw	lwnpu
8	hzzo	hz	vaozm	ocz	ojbv	kvmot
9	gyyn	gy	uznyl	nby	niau	julns
10	fxxm	fx	tymxk	max	mhzt	itkmr
11	ewwl	ew	sxlwj	lzw	lgys	hsjlg
12	dvvk	dv	rwkvi	kyv	kfxr	grikp
13	cuuj	cu	qvjuh	jxu	jewq	fghjo
14	btti	bt	puitg	iwt	idvp	epgin
15	assh	as	othsf	hvs	hcuo	dofhm
16	zrrg	zr	nsgre	gur	gbtn	cnegl
17	yqqf	yq	mrfqd	ftq	fasm	bmghk
18	xppe	xp	lqepc	esp	ezrl	alcej
19	wood	wo	kpdob	dro	dyqk	zkbdi
20	vnnc	vn	jocna	cqn	cxpj	yjach
21	ummb	um	inbmz	bpm	bwoi	xizbg
22	tlla	tl	hmaly	aol	avnh	whyaf
23	skkz	sk	glzxx	znk	zumg	vgxze
24	rjjy	rj	fkyjw	ymj	ytlf	ufwyd
25	qiix	qi	ejxiv	xli	xske	tevxv

Monoalphabetic Cipher

- ❑ rather than just shifting the alphabet
- ❑ could shuffle (jumble) the letters arbitrarily
- ❑ each plaintext letter maps to a different random ciphertext letter
- ❑ hence key is 26 letters long

Plain: abcdefghijklmnopqrstuvwxyz

Cipher: DKVQFIBJWPESCXHTMYAUOLRGZN

Plaintext: ifwewishtoreplaceletters

Ciphertext: WIRFRWAJUHYFTSDVFSFUUFYA

With only 25 possible keys, the Caesar cipher is far from secure. A dramatic increase in the key space can be achieved by allowing an arbitrary substitution, where the translation alphabet can be any permutation of the 26 alphabetic characters. A permutation of a finite set of elements S is an ordered sequence of all the elements of S , with each element appearing exactly once. In general, there are $n!$ permutations of a set of n elements.

See text example of a translation alphabet, and an encrypted message using it.

Monoalphabetic Cipher Security

- ❑ now have a total of $26! = 4 \times 10^{26}$ keys
- ❑ with so many keys, might think is secure
- ❑ but would be **!!!WRONG!!!**
- ❑ problem is language characteristics

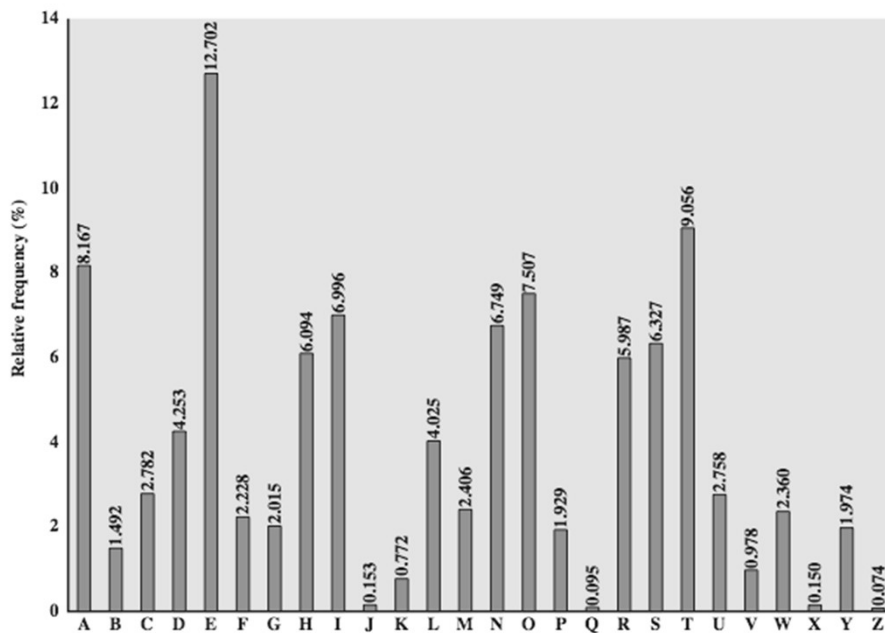
Note that even given the very large number of keys, being 10 orders of magnitude greater than the key space for DES, the monoalphabetic substitution cipher is not secure, because it does not sufficiently obscure the underlying language characteristics.

Language Redundancy and Cryptanalysis

- human languages are **redundant**
- eg "how are you" "how r u"
- letters are not equally commonly used
- in English E is by far the most common letter
 - followed by T,R,N,I,O,A,S
- other letters like Z,J,K,Q,X are fairly rare
- have tables of single, double & triple letter frequencies for various languages

As the example shows, we don't actually need all the letters in order to understand written English text. Here vowels were removed, but they're not the only redundancy. cf written Hebrew has no vowels for same reason. Are usually familiar with "party conversations", can hear one person speaking out of hubbub of many, again because of redundancy in aural language also. This redundancy is also the reason we can compress text files, the computer can derive a more compact encoding without losing any information. Basic idea is to count the relative frequencies of letters, and note the resulting pattern.

English Letter Frequencies



Note that all human languages have varying letter frequencies, though the number of letters and their frequencies varies. Stallings Figure 2.5 shows English letter frequencies. Seberry & Pieprzyk, "Cryptography - An Introduction to Computer Security", Prentice-Hall 1989, Appendix A has letter frequency graphs for 20 languages (most European & Japanese & Malay). Also useful are tables of common two-letter combinations, known as digrams, and three-letter combinations, known as trigrams.

Use in Cryptanalysis

- ❑ key concept - monoalphabetic substitution ciphers do not change relative letter frequencies
- ❑ discovered by Arabian scientists in 9th century
- ❑ calculate letter frequencies for ciphertext
- ❑ compare counts/plots against known values
- ❑ if caesar cipher look for common peaks/troughs
 - peaks at: A-E-I triple, NO pair, RST triple
 - troughs at: JK, X-Z
- ❑ for monoalphabetic must identify each letter
 - tables of common double/triple letters help

The simplicity and strength of the monoalphabetic substitution cipher meant it dominated cryptographic use for the first millenium AD. It was broken by Arabic scientists. The earliest known description is in Abu al-Kindi's "A Manuscript on Deciphering Cryptographic Messages", published in the 9th century but only rediscovered in 1987 in Istanbul, but other later works also attest to their knowledge of the field. Monoalphabetic ciphers are easy to break because they reflect the frequency data of the original alphabet. The cryptanalyst looks for a mapping between the observed pattern in the ciphertext, and the known source language letter frequencies. If English, look for peaks at: A-E-I triple, NO pair, RST triple, and troughs at: JK, X-Z.

Monoalphabetic ciphers are easy to break because they reflect the frequency data of the original alphabet.

Example Cryptanalysis

□ given ciphertext:

```
UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
VUEPHZHMZDZSHZOWSFPAPPDTSVPPQUZWMYXUZUHSX
EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ
```

□ count relative letter frequencies (see text)

□ guess P & Z are e and t

□ guess ZW is th and hence ZWP is the

□ proceeding with trial and error finally get:

```
it was disclosed yesterday that several informal but
direct contacts have been made with political
representatives of the viet cong in moscow
```

Illustrate the process with this example from the text in Stallings section 2.2. Comparing letter frequency breakdown with Figure 2.5, it seems likely that cipher letters P and Z are the equivalents of plain letters e and t, but it is not certain which is which. The letters S, U, O, M, and H are all of relatively high frequency and probably correspond to plain letters from the set {a, h, i, n, o, r, s}. The letters with the lowest frequencies (namely, A, B, G, Y, I, J) are likely included in the set {b, j, k, q, v, x, z}. A powerful tool is to look at the frequency of two-letter combinations, known as digrams. A table similar to Figure 2.5 could be drawn up showing the relative frequency of digrams. The most common such digram is th. In our ciphertext, the most common digram is ZW, which appears three times. So we make the correspondence of Z with t and W with h. Then, by our earlier hypothesis, we can equate P with e. Now notice that the sequence ZWP appears in the ciphertext, and we can translate that sequence as "the." This is the most frequent trigram (three-letter combination) in English, which seems to indicate that we are on the right track. Next, notice the sequence ZWSZ in the first line. We do not know that these four letters form a complete word, but if they do, it is of the form th_t. If so, S equates with a. Only four letters have been identified, but already we have quite a bit of the message. Continued analysis of frequencies plus trial and error should easily yield a solution from this point. The complete plaintext, with spaces added between words, is shown on slide.

Playfair Cipher

- not even the large number of keys in a monoalphabetic cipher provides security
- one approach to improving security was to encrypt multiple letters
- the **Playfair Cipher** is an example
- invented by Charles Wheatstone in 1854, but named after his friend Baron Playfair

Consider ways to reduce the "spikyness" of natural language text, since if just map one letter always to another, the frequency distribution is just shuffled. One approach is to encrypt more than one letter at once. The Playfair cipher is an example of doing this, treats digrams in the plaintext as single units and translates these units into ciphertext digrams.

Playfair Key Matrix

- a 5X5 matrix of letters based on a keyword
- fill in letters of keyword (sans duplicates)
- fill rest of matrix with other letters
- eg. using the keyword MONARCHY

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

The best-known multiple-letter encryption cipher is the Playfair, which treats digrams in the plaintext as single units and translates these units into ciphertext digrams. The Playfair algorithm is based on the use of a 5x5 matrix of letters constructed using a keyword. The rules for filling in this 5x5 matrix are: L to R, top to bottom, first with keyword after duplicate letters have been removed, and then with the remain letters, with I/J used as a single letter. This example comes from Dorothy Sayer's book "Have His Carcase", in which Lord Peter Wimsey solves it, and describes the use of a probably word attack.

Encrypting and Decrypting

- plaintext is encrypted two letters at a time
 1. if a pair is a repeated letter, insert filler like 'X'
 2. if both letters fall in the same row, replace each with letter to right (wrapping back to start from end)
 3. if both letters fall in the same column, replace each with the letter below it (wrapping to top from bottom)
 4. otherwise each letter is replaced by the letter in the same row and in the column of the other letter of the pair

Plaintext is encrypted two letters at a time, according to the rules as shown. Note how you wrap from right side back to left, or from bottom back to top.

1. if a pair is a repeated letter, insert a filler like 'X', eg. "balloon" encrypts as "ba lx lo on"
2. if both letters fall in the same row, replace each with letter to right (wrapping back to start from end), eg. "ar" encrypts as "RM"
3. if both letters fall in the same column, replace each with the letter below it (again wrapping to top from bottom), eg. "mu" encrypts to "CM"
4. otherwise each letter is replaced by the one in its row in the column of the other letter of the pair, eg. "hs" encrypts to "BP", and "ea" to "IM" or "JM" (as desired)

Decrypting of course works exactly in reverse. Can see this by working the example pairs shown, backwards.

Security of Playfair Cipher

- security much improved over monoalphabetic
- since have $26 \times 26 = 676$ digrams
- would need a 676 entry frequency table to analyse (verses 26 for a monoalphabetic)
- and correspondingly more ciphertext
- was widely used for many years
 - eg. by US & British military in WW1
- it **can** be broken, given a few hundred letters
- since still has much of plaintext structure

The Playfair cipher is a great advance over simple monoalphabetic ciphers, since there are $26 \times 26 = 676$ digrams (vs 26 letters), so that identification of individual digrams is more difficult. Also, the relative frequencies of individual letters exhibit a much greater range than that of digrams, making frequency analysis much more difficult. The Playfair cipher was for a long time considered unbreakable. It was used as the standard field system by the British Army in World War I and still enjoyed considerable use by the U.S. Army and other Allied forces during World War II. Despite this level of confidence in its security, the Playfair cipher is relatively easy to break because it still leaves much of the structure of the plaintext language intact. A few hundred letters of ciphertext are generally sufficient.

Polyalphabetic Ciphers

- **polyalphabetic substitution ciphers**
- improve security using multiple cipher alphabets
- make cryptanalysis harder with more alphabets to guess and flatter frequency distribution
- use a key to select which alphabet is used for each letter of the message
- use each alphabet in turn
- repeat from start after end of key is reached

One approach to reducing the "spikyness" of natural language text is used the Playfair cipher which encrypts more than one letter at once. We now consider the other alternative, using multiple cipher alphabets in turn. This gives the attacker more work, since many alphabets need to be guessed and because the frequency distribution is more complex, since the same plaintext letter could be replaced by several ciphertext letters, depending on which alphabet is used. The general name for this approach is a polyalphabetic substitution cipher. All these techniques have the following features in common:

1. A set of related monoalphabetic substitution rules is used.
2. A key determines which particular rule is chosen for a given transformation.

Vigenère Cipher

- ❑ simplest polyalphabetic substitution cipher
- ❑ effectively multiple caesar ciphers
- ❑ key is multiple letters long $K = k_1 k_2 \dots k_d$
- ❑ i^{th} letter specifies i^{th} alphabet to use
- ❑ use each alphabet in turn
- ❑ repeat from start after d letters in message
- ❑ decryption simply works in reverse

The best known, and one of the simplest, such algorithms is referred to as the Vigenère cipher, where the set of related monoalphabetic substitution rules consists of the 26 Caesar ciphers, with shifts of 0 through 25. Each cipher is denoted by a key letter, which is the ciphertext letter that substitutes for the plaintext letter 'a', and which are each used in turn, as shown next.

Example of Vigenère Cipher

- write the plaintext out
- write the keyword repeated above it
- use each key letter as a caesar cipher key
- encrypt the corresponding plaintext letter
- eg using keyword *deceptive*

key:

deceptivedeceptivedeceptive

plaintext: wearediscoveredsaveyourself

ciphertext: ZICVTWQNGRZGVTWAVZHCQYGLMGJ

Discuss this simple example from text Stallings section 2.2.

Example of Vigenère Cipher

key: *deceptivedeceptivedeceptive*
plaintext: *wearediscoveredsaveyourself*
ciphertext: *ZICVTWQNGRZGVTWAVZHCQYGLMGJ*

Expressed numerically, we have the following result.

key	3	4	2	4	15	19	8	21	4	3	4	2	4	15
plaintext	22	4	0	17	4	3	8	18	2	14	21	4	17	4
ciphertext	25	8	2	21	19	22	16	13	6	17	25	6	21	19

key	19	8	21	4	3	4	2	4	15	19	8	21	4
plaintext	3	18	0	21	4	24	14	20	17	18	4	11	5
ciphertext	22	0	21	25	7	2	16	24	6	11	12	6	9

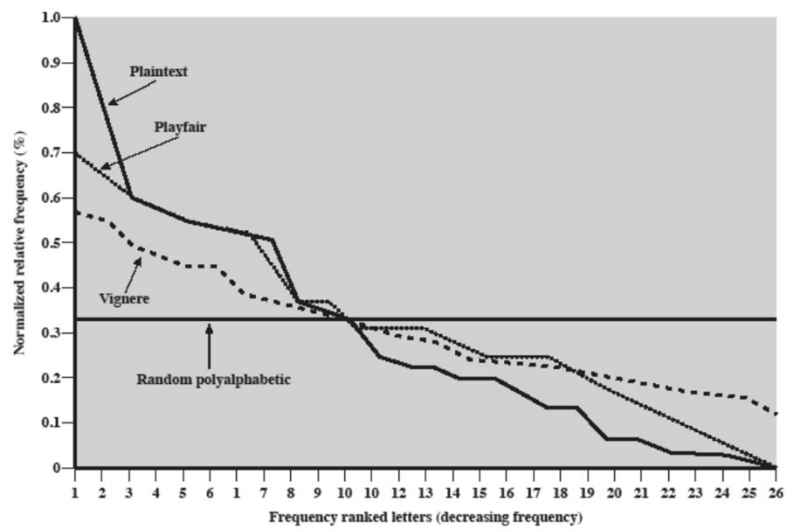


Figure 2.6 Relative Frequency of Occurrence of Letters

Aids

- ❑ simple aids can assist with en/decryption
- ❑ a **Saint-Cyr Slide** is a simple manual aid
 - a slide with repeated alphabet
 - line up plaintext 'A' with key letter, eg 'C'
 - then read off any mapping for key letter
- ❑ can bend round into a **cipher disk**
- ❑ or expand into a **Vigenère Tableau**

Implementing polyalphabetic ciphers by hand can be very tedious. Various aids were devised to assist the process.

The "*Saint-Cyr Slide*" was popularised and named by Jean Kerckhoffs, who published a famous early text "*La Cryptographie Militaire*" (Military Cryptography) in 1883. He named the slide after the French National Military Academy where the methods were taught. He also noted that any slide can be expanded into a tableau, or bent round into a cipher disk.

The *Vigenère Tableau* (given in Stallings 4/e as Table 2.3) is a complete set of forward shifted alphabet mappings.

Security of Vigenère Ciphers

- ❑ have multiple ciphertext letters for each plaintext letter
- ❑ hence letter frequencies are obscured
- ❑ but not totally lost
- ❑ start with letter frequencies
 - see if look monoalphabetic or not
- ❑ if not, then need to determine number of alphabets, since then can attach each

The Vigenère & related polyalphabetic ciphers still do not completely obscure the underlying language characteristics. The strength of this cipher is that there are multiple ciphertext letters for each plaintext letter, one for each unique letter of the keyword. Thus, the letter frequency information is obscured. However, not all knowledge of the plaintext structure is lost. The key to breaking them is to identify the number of translation alphabets, and then attack each separately. If a monoalphabetic substitution is used, then the statistical properties of the ciphertext should be the same as that of the language of the plaintext. If, on the other hand, a Vigenère cipher is suspected, then progress depends on determining the length of the keyword.

Kasiski Method

- ❑ method developed by Babbage / Kasiski
- ❑ repetitions in ciphertext give clues to period
- ❑ so find same plaintext an exact period apart
- ❑ which results in the same ciphertext
- ❑ of course, could also be random fluke
- ❑ eg repeated "VTW" in previous example
- ❑ suggests size of 3 or 9
- ❑ then attack each monoalphabetic cipher individually using same techniques as before

For some centuries the Vigenère cipher was *le chiffre indéchiffrable* (the unbreakable cipher). As a result of a challenge, it was broken by Charles Babbage (the inventor of the computer) in 1854 but kept secret (possibly because of the Crimean War - not the first time governments have kept advances to themselves!). The method was independently reinvented by a Prussian, Friedrich Kasiski, who published the attack now named after him in 1863. However lack of major advances meant that various polyalphabetic substitution ciphers were used into the 20C. One very famous incident was the breaking of the Zimmermann telegram in WW1 which resulted in the USA entering the war. The important is that if two identical sequences of plaintext letters occur at a distance that is an integer multiple of the keyword length, they will generate identical ciphertext sequences.

In general the approach is to find a number of duplicated sequences, collect all their distances apart, look for common factors, remembering that some will be random flukes and need to be discarded. Now have a series of monoalphabetic ciphers, each with original language letter frequency characteristics. Can attack these in turn to break the cipher.

Autokey Cipher

- ❑ ideally want a key as long as the message
- ❑ Vigenère proposed the **autokey** cipher
- ❑ with keyword is prefixed to message as key
- ❑ knowing keyword can recover the first few letters
- ❑ use these in turn on the rest of the message
- ❑ but still have frequency characteristics to attack
- ❑ eg. given key *deceptive*

```
key:      deceptive
plaintext: wearediscoveredsaveyourself
ciphertext: ZICVTWQNGKZEIIGASXSTSLVVWLA
```

Taking the polyalphabetic idea to the extreme, want as many different translation alphabets as letters in the message being sent. One way of doing this with a smallish key, is to use the Autokey cipher.

The example uses the keyword "DECEPTIVE" prefixed to as much of the message "WEAREDISCOVEREDSAV" as is needed. When deciphering, recover the first 9 letters using the keyword "DECEPTIVE". Then instead of repeating the keyword, start using the recovered letters from the message "WEAREDISC". As recover more letters, have more of key to recover later letters.

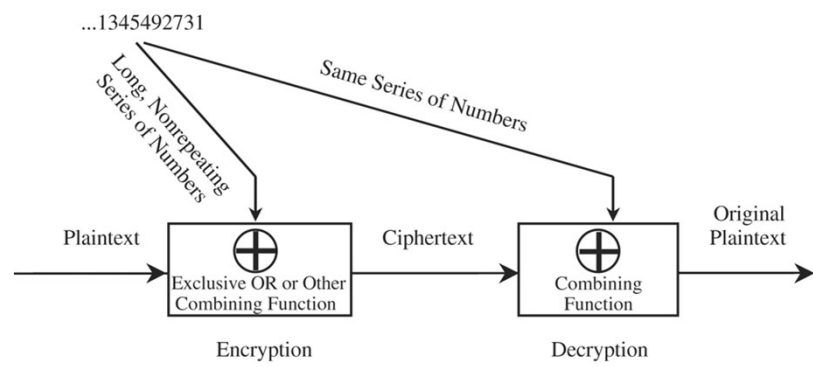
Problem is that the same language characteristics are used by the key as the message. ie. a key of 'E' will be used more often than a 'T' etc hence an 'E' encrypted with a key of 'E' occurs with probability $(0.1275)^2 = 0.01663$, about twice as often as a 'T' encrypted with a key of 'T' have to use a larger frequency table, but it exists given sufficient ciphertext this can be broken.

Vernam Cipher

- ultimate defense is to use a key as long as the plaintext
- with no statistical relationship to it
- invented by AT&T engineer Gilbert Vernam in 1918
- originally proposed using a very long but eventually repeating key

The ultimate defense against such a cryptanalysis is to choose a keyword that is as long as the plaintext and has no statistical relationship to it. Such a system was introduced by an AT&T engineer named Gilbert Vernam in 1918. His system works on binary data (bits) rather than letters. The system can be expressed succinctly as follows: $c_i = p_i \text{ XOR } k_i$

The essence of this technique is the means of construction of the key. Vernam proposed the use of a running loop of tape that eventually repeated the key, so that in fact the system worked with a very long but repeating keyword. Although such a scheme, with a long key, presents formidable cryptanalytic difficulties, it can be broken with sufficient ciphertext, the use of known or probable plaintext sequences, or both.



Vernam Cipher.

One-Time Pad

- ❑ if a truly random key as long as the message is used, the cipher will be secure
- ❑ called a One-Time pad
- ❑ is unbreakable since ciphertext bears no statistical relationship to the plaintext
- ❑ since for **any plaintext & any ciphertext** there exists a key mapping one to other
- ❑ can only use the key **once** though
- ❑ problems in generation & safe distribution of key

The One-Time Pad is an evolution of the Vernham cipher. An Army Signal Corp officer, Joseph Mauborgne, proposed an improvement using a random key that was truly as long as the message, with no repetitions, which thus totally obscures the original message. It produces random output that bears no statistical relationship to the plaintext. Because the ciphertext contains no information whatsoever about the plaintext, there is simply no way to break the code, since any plaintext can be mapped to any ciphertext given some key.

The one-time pad offers complete security but, in practice, has two fundamental difficulties:

1. There is the practical problem of making large quantities of random keys.
2. And the problem of key distribution and protection, where for every message to be sent, a key of equal length is needed by both sender and receiver.

Because of these difficulties, the one-time pad is of limited utility, and is useful primarily for low-bandwidth channels requiring very high security. The one-time pad is the only cryptosystem that exhibits what is referred to as *perfect secrecy*.

Transposition Ciphers

- now consider classical **transposition** or **permutation** ciphers
- these hide the message by rearranging the letter order
- without altering the actual letters used
- can recognise these since have the same frequency distribution as the original text

All the techniques examined so far involve the substitution of a ciphertext symbol for a plaintext symbol. A very different kind of mapping is achieved by performing some sort of permutation on the plaintext letters. This technique is referred to as a transposition cipher, and form the second basic building block of ciphers. The core idea is to rearrange the order of basic units (letters/bytes/bits) without altering their actual values.

Rail Fence cipher

- ❑ write message letters out diagonally over a number of rows
- ❑ then read off cipher row by row
- ❑ eg. write message out as:

m e m a t r h t g p r y
e t e f e t e o a a t

- ❑ giving ciphertext

MEMATRHTGPRYETEFETEOAAT

The simplest such cipher is the rail fence technique, in which the plaintext is written down as a sequence of diagonals and then read off as a sequence of rows.

The example message is: "meet me after the toga party" with a rail fence of depth 2.

This sort of thing would be trivial to cryptanalyze.

Row Transposition Ciphers

- is a more complex transposition
- write letters of message out in rows over a specified number of columns
- then reorder the columns according to some key before reading off the rows

Key: 4312567

Column Out 4 3 1 2 5 6 7

Plaintext: a t t a c k p

o s t p o n e

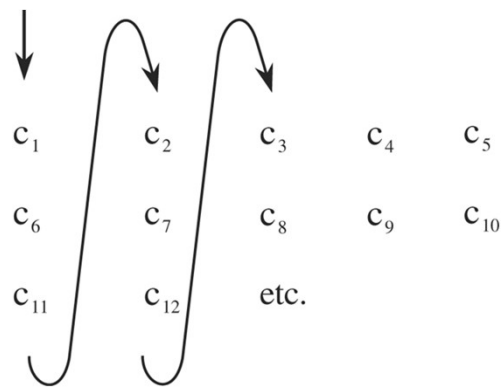
d u n t i l t

w o a m x y z

Ciphertext: TTNAAPTMTSUOAODWCOIXKNLYPETZ

A more complex transposition cipher is to write the message in a rectangle, row by row, and read the message off shuffling the order of the columns in each row. The order of the columns then becomes the key to the algorithm. In the example shown, the key is 4312567, that is use column 4 first, then column 3, then 1 etc (as shown in the Column Out row).

A pure transposition cipher is easily recognized because it has the same letter frequencies as the original plaintext. For the type of columnar transposition just shown, cryptanalysis is fairly straightforward and involves laying out the ciphertext in a matrix and playing around with column positions. Digram and trigram frequency tables can be useful.



Columnar Transposition.

Product Ciphers

- ❑ ciphers using substitutions or transpositions are not secure because of language characteristics
- ❑ hence consider using several ciphers in succession to make harder, but:
 - two substitutions make a more complex substitution
 - two transpositions make more complex transposition
 - but a substitution followed by a transposition makes a new much harder cipher
- ❑ this is bridge from classical to modern ciphers

Have seen that ciphers based on just substitutions or transpositions are not secure, and can be attacked because they do not sufficient obscure the underlying language structure

So consider using several ciphers in succession to make harder.

A substitution followed by a transposition is known as a Product Cipher, and makes a new much more secure cipher, and forms the bridge to modern ciphers.

Rotor Machines

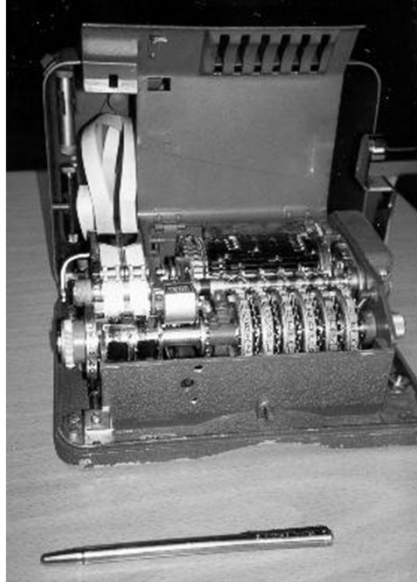
- ❑ before modern ciphers, rotor machines were most common complex ciphers in use
- ❑ widely used in WW2
 - German Enigma, Allied Hagelin, Japanese Purple
- ❑ implemented a very complex, varying substitution cipher
- ❑ used a series of cylinders, each giving one substitution, which rotated and changed after each letter was encrypted
- ❑ with 3 cylinders have $26^3=17576$ alphabets

The next major advance in ciphers required use of mechanical cipher machines which enabled to use of complex varying substitutions.

A rotor machine consists of a set of independently rotating cylinders through which electrical pulses can flow. Each cylinder has 26 input pins and 26 output pins, with internal wiring that connects each input pin to a unique output pin. If we associate each input and output pin with a letter of the alphabet, then a single cylinder defines a monoalphabetic substitution. After each input key is depressed, the cylinder rotates one position, so that the internal connections are shifted accordingly. The power of the rotor machine is in the use of multiple cylinders, in which the output pins of one cylinder are connected to the input pins of the next, and with the cylinders rotating like an “odometer”, leading to a very large number of substitution alphabets being used, eg with 3 cylinders have $26^3=17576$ alphabets used.

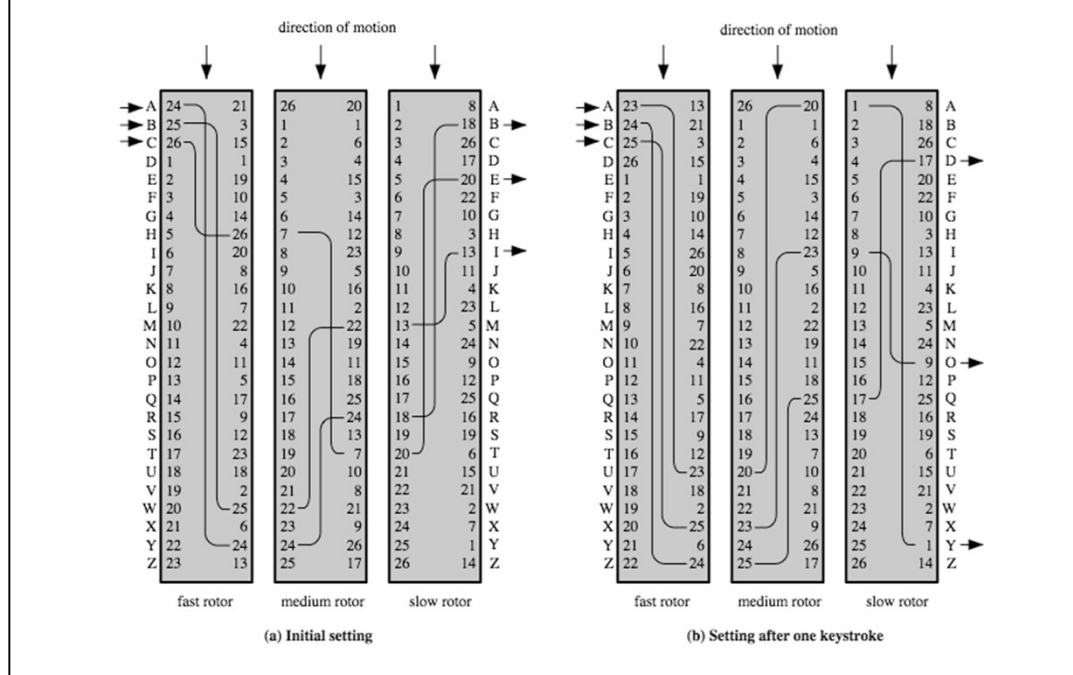
They were extensively used in world war 2, and the history of their use and analysis is one of the great stories from WW2.

Hagelin Rotor Machine



This photo of an Allied *Hagelin* machine was taken by Lawrie Brown at Eurocrypt'93 in Norway. Note pen for scale, and the rotating cipher wheels near the front.

Rotor Machine Principles



The basic principle of the rotor machine is illustrated in Figure 2.8. The machine consists of a set of independently rotating cylinders through which electrical pulses can flow. Each cylinder has 26 input pins and 26 output pins, with internal wiring that connects each input pin to a unique output pin. If we associate each input and output pin with a letter of the alphabet, then a single cylinder defines a monoalphabetic substitution. If an operator depresses the key for the letter A, an electric signal is applied to the first pin of the first cylinder and flows through the internal connection to the twenty-fifth output pin. Consider a machine with a single cylinder. After each input key is depressed, the cylinder rotates one position, so that the internal connections are shifted accordingly. Thus, a different monoalphabetic substitution cipher is defined. After 26 letters of plaintext, the cylinder would be back to the initial position. Thus, we have a polyalphabetic substitution algorithm with a period of 26.

A single-cylinder system is trivial and does not present a formidable cryptanalytic task. The power of the rotor machine is in the use of multiple cylinders, in which the output pins of one cylinder are connected to the input pins of the next. Figure 2.8 shows a three-cylinder system. With multiple cylinders, the one closest to the operator input rotates one pin position with each keystroke. The right half of Figure 2.8 shows the system's configuration after a single keystroke. For every complete rotation of the inner cylinder, the middle cylinder rotates one pin position. Finally, for every complete rotation of the middle cylinder, the outer cylinder rotates one pin position. The result is that there are $26 \times 26 \times 26 = 17,576$ different substitution alphabets used before the system repeats.

Steganography

- ❑ an alternative to encryption
- ❑ hides existence of message
 - using only a subset of letters/words in a longer message marked in some way
 - using invisible ink
 - hiding in LSB in graphic image or sound file
- ❑ has drawbacks
 - high overhead to hide relatively few info bits
- ❑ advantage is can obscure encryption use

Steganography is an alternative to encryption which hides the very existence of a message by some means. There are a large range of techniques for doing this.

Steganography has a number of drawbacks when compared to encryption. It requires a lot of overhead to hide a relatively few bits of information.

Also, once the system is discovered, it becomes virtually worthless, although a message can be first encrypted and then hidden using steganography. The advantage of steganography is that it can be employed by parties who have something to lose should the fact of their secret communication (not necessarily the content) be discovered.

Summary

□ have considered:

- classical cipher techniques and terminology
- monoalphabetic substitution ciphers
- cryptanalysis using letter frequencies
- Playfair cipher
- polyalphabetic ciphers
- transposition ciphers
- product ciphers and rotor machines
- stenography

Chapter 2 summary.